



ITI Cmos Analog IC Design
Lab 11
Fully-Differential Folded Cascode OTA
Mini project 2

Student name: Mohamed Adel Ali

Section 7 online

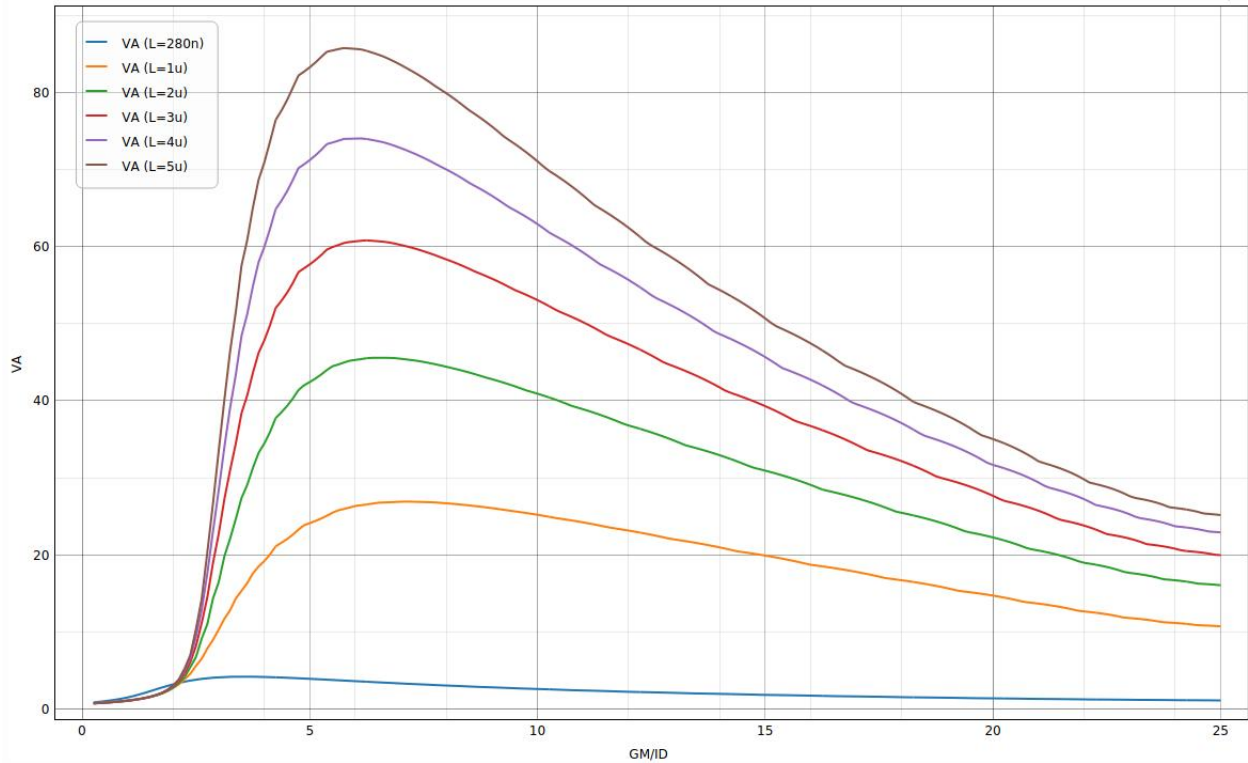
Specifications

Parameter	Required
Technology	0.13um GF180MCU
Supply voltage	2.5V
Closed loop gain	2
Phase margin @ ACL	$\geq 70^\circ$
CM input range – low	≤ 0 V
CM input range – high	≥ 1 V
Differential output swing	1.2 V _{pk-pk}
Load capacitance	500 fF
DC loop gain	60 dB
CL settling time (1% error)	100 ns

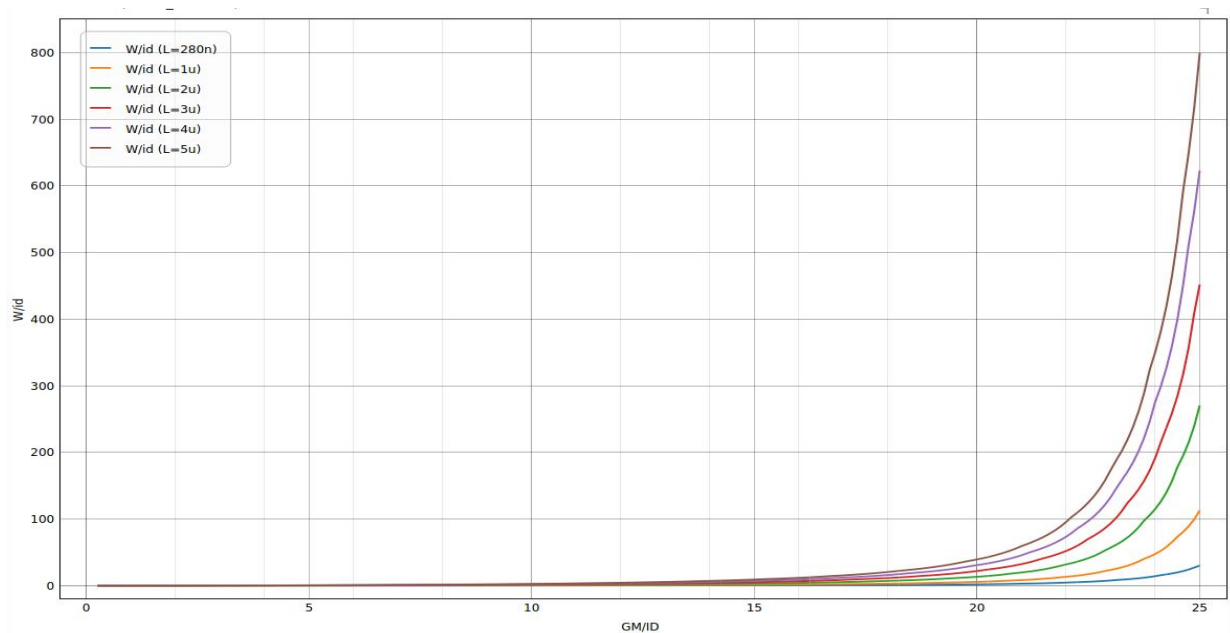
Part 1 – gm/ID Design Charts

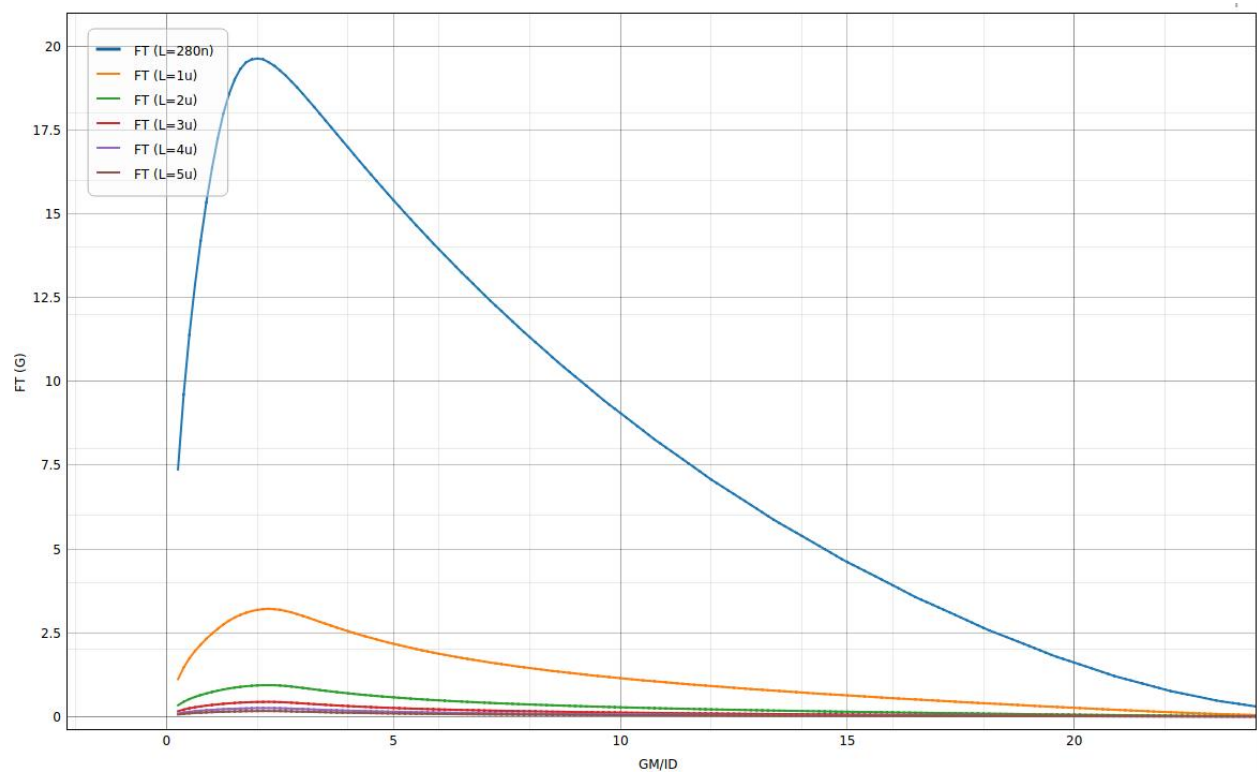
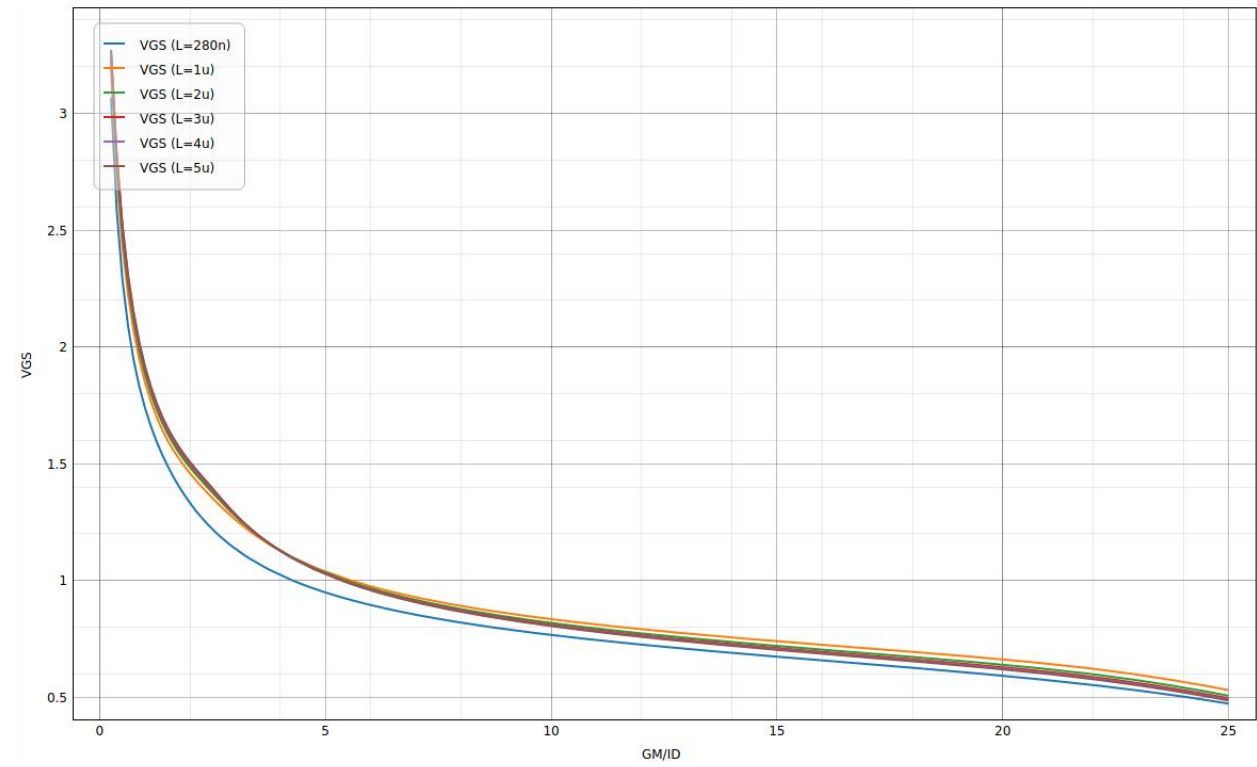
NMOS Charts

Early Voltage (VA) vs gm/ID



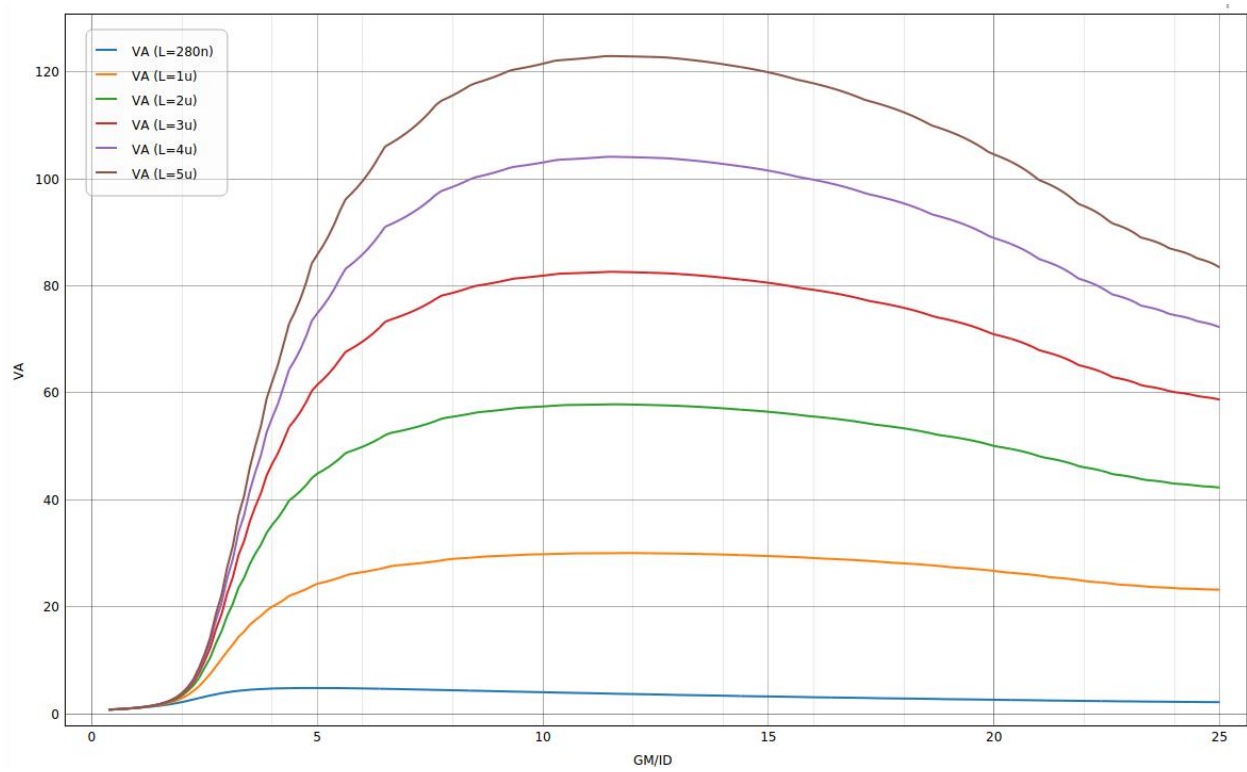
W/ID vs gm/ID



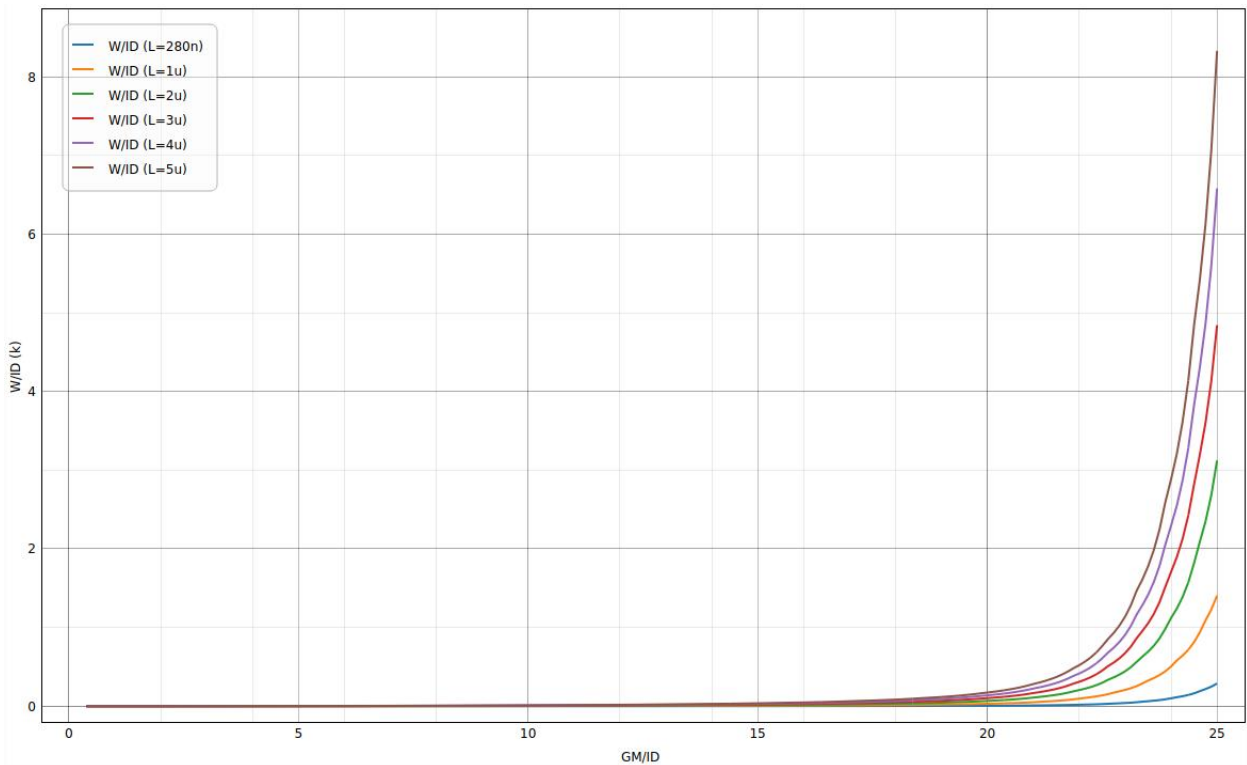
fT vs gm/ID**VGS vs gm/ID**

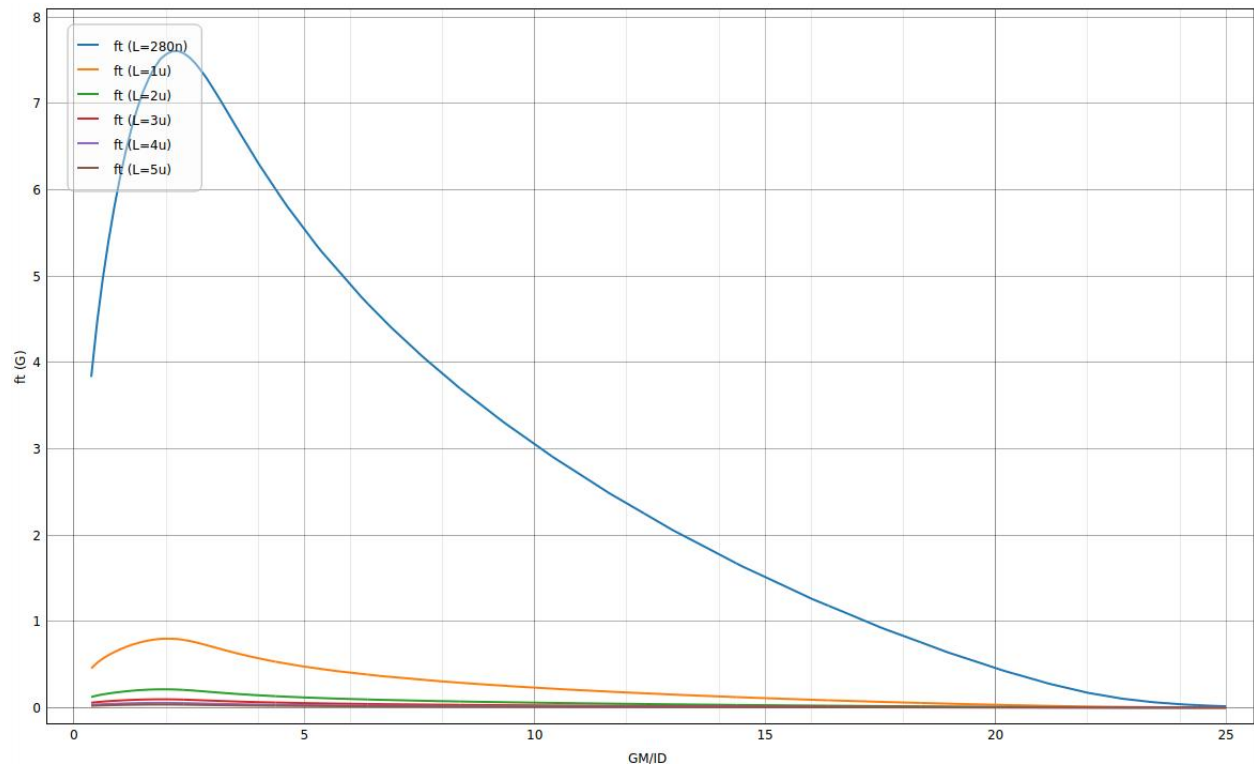
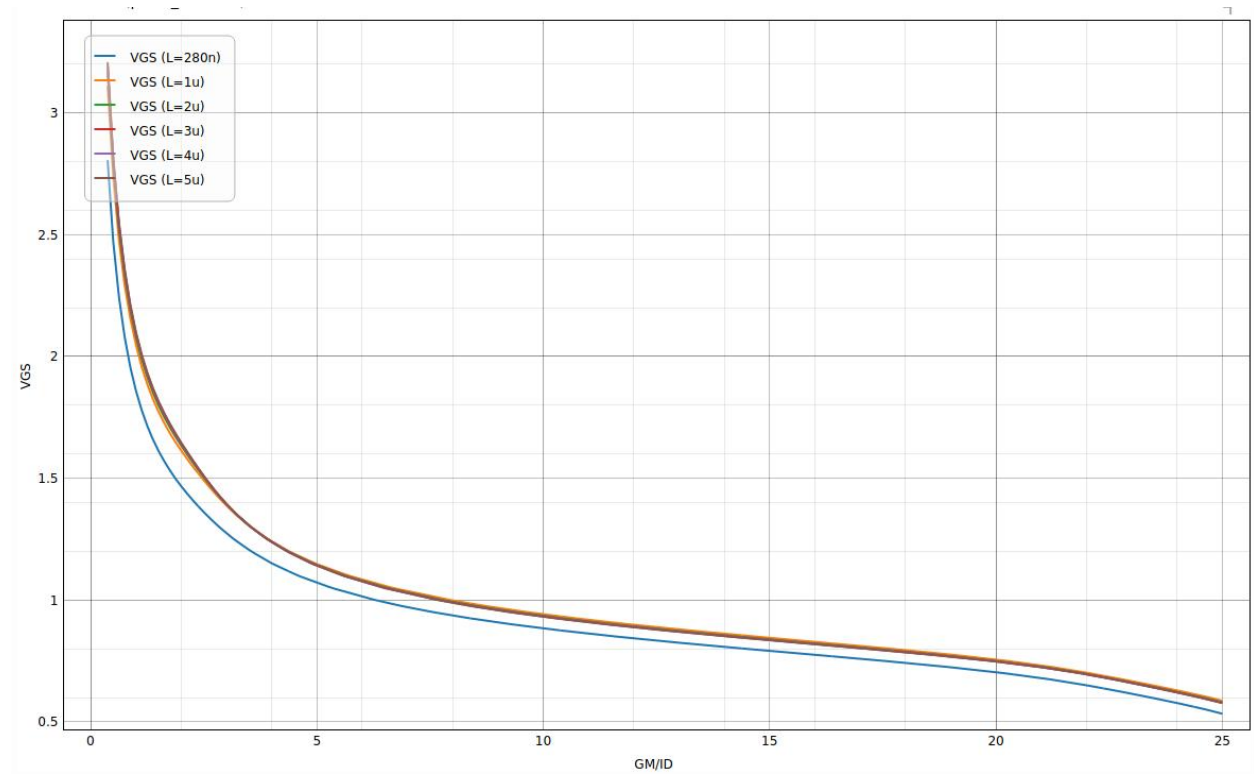
PMOS Charts

Early Voltage (VA) vs gm/ID



W/ID vs gm/ID



fT vs gm/ID**VGS vs gm/ID**

Part 2 – OTA Hand Design

From the CMIR specification a PMOS input stage is required (input pair biased in moderate/weak inversion for high GBW efficiency and low input capacitance).

Design Assumptions

Device group	L	gm/ID
Input pair (pmos)	300n	17
Current source	1u	10
Cascode	500n	15(before tuning)

Hand Calculation Steps

1) Closed-loop bandwidth from settling-time spec

Convert the 1% settling-time spec to closed-loop bandwidth:

$$t_s(1\%) \approx 4.6 / \omega_{BW}$$

$$f_{BW,CL} = 7.33 \text{ Mhz} \quad GBW = 3 * \omega_{BW,CL} = 22 \text{ Mhz}$$

2) Bias current from bandwidth spec

$$G_{m1,2} = 2 * GBW * \pi * C_{tot} = 170 \text{ uS} \quad , C_{tot} = C_L + C_F + C_{in}$$

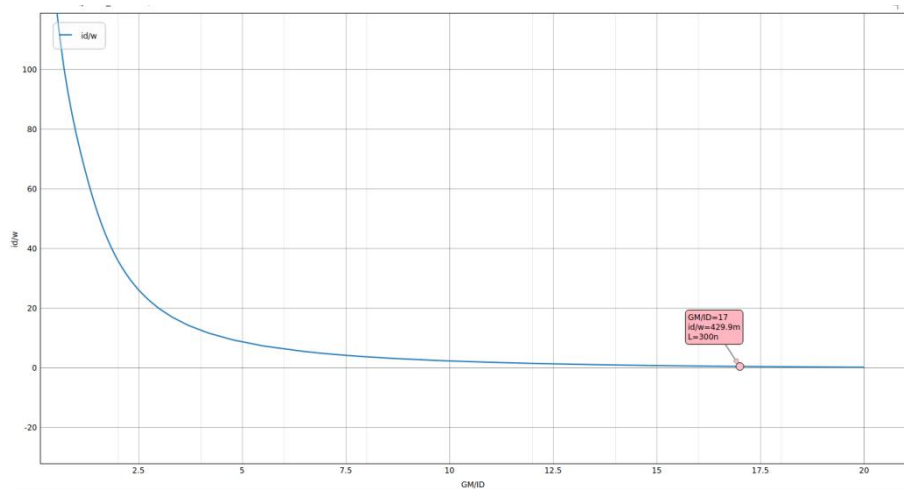
$$I_d (\text{input pair}) = 170\text{u} / 17 = 10\text{u}$$

$$I_D (\text{input pair, per branch}) = 10\text{u}$$

$$I_D (\text{cascode branch}) = 5 \text{ u} \quad (\text{making current in input pair } 2 * \text{current of cascode branch})$$

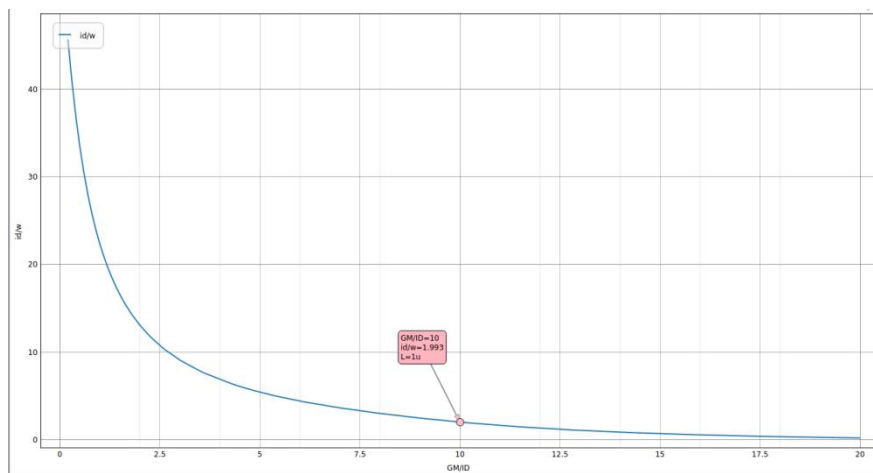
$$I_D (\text{NMOS current source}) = 15\text{u} \quad (i_{b1} + i_{b2})$$

- input pair
from chart id/w vs gm/id



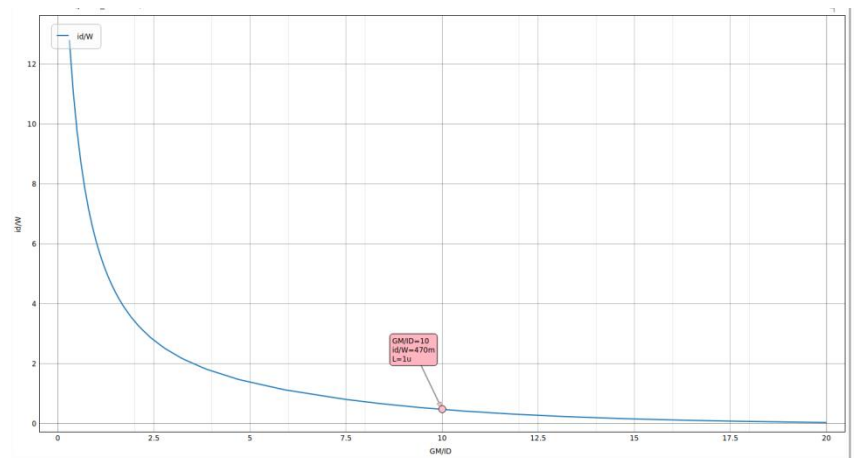
$$W = id/w / id = 23.2u$$

-Current source (NMOS)
from chart id/w vs gm/id



$$W = id/w / id = 7.526 \text{ for } 15 \text{ u}$$

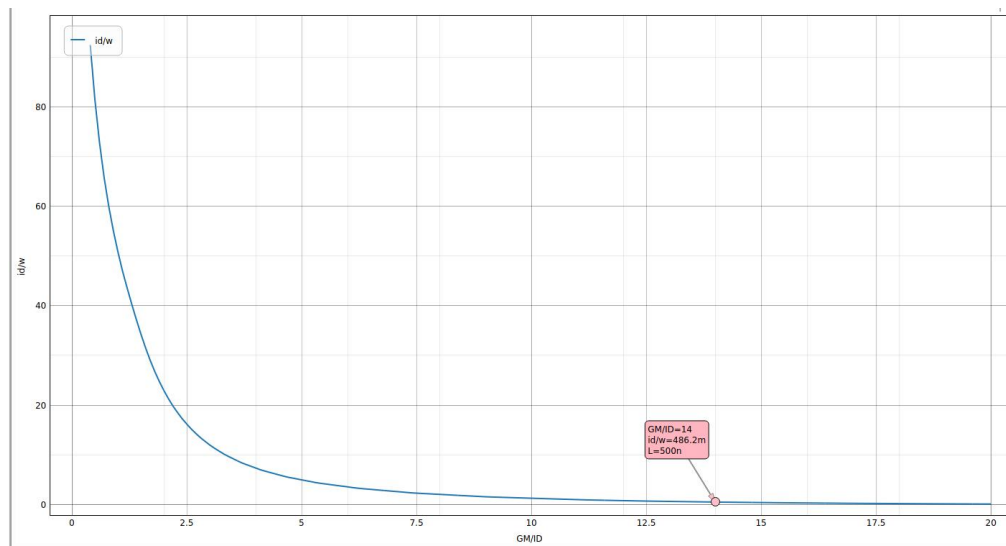
- Current source (PMOS)
from chart i_d/w vs g_m/i_d



$$W = i_d/w / i_d = 10.6 \text{ for } 5 \mu$$

- Cascode (PMOS)

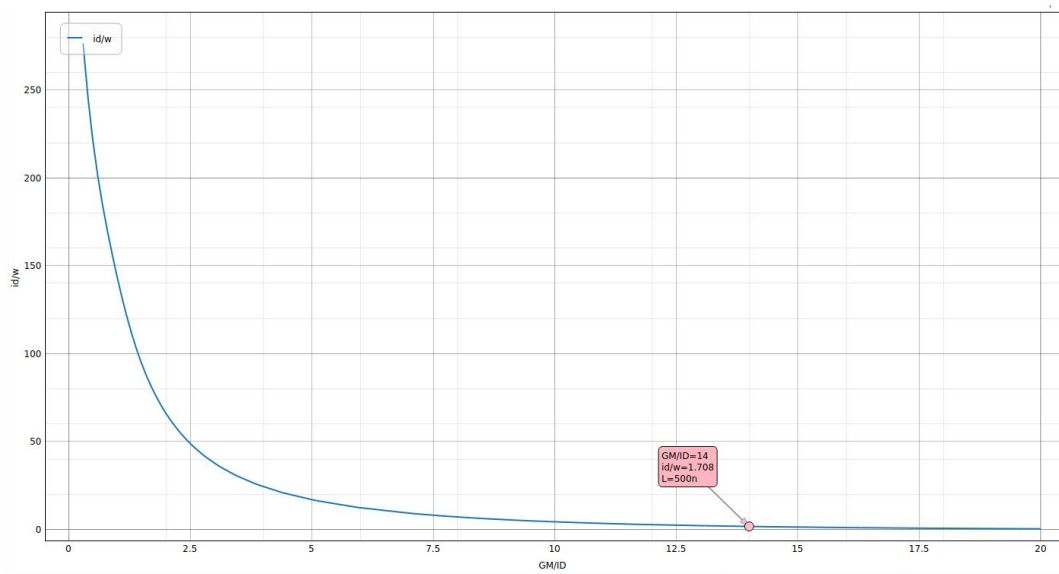
Choosing $g_m/i_d = 14$ instead of 15 to decrease g_m/g_{ds}
from chart i_d/w vs g_m/i_d



$$W = i_d/w / i_d = 10.6 \text{ for}$$

- Cascode (NMOS)

Choosing $gm/id = 14$ instead of 15 to decrease gm/g_{ds} and overall gain of OTA from chart id/w vs gm/id



$$W = id/w / id = 2.9\mu$$

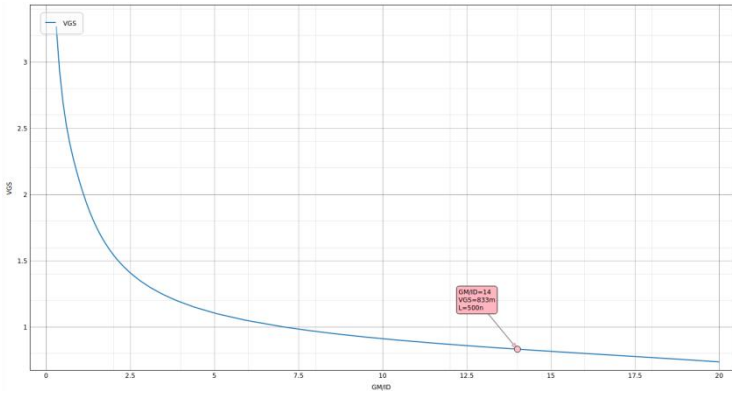
3) Transistor sizing (W) from the gm/ID charts

Transistor	Type	gm/ID (1/V)	ID (uA)	L (um)	W (um)
M0 , M1	PMOS-input pair	17	10u	300n	23.2u
M2	PMOS-Current source	10	20u	1u	42.4u
M3,M4	PMOS-Current source	10	5u	1u	10.6u
M5,M6	PMOS-Cascode	14	5u	500n	10.6u
M7,M8	NMOS-Cascode	14	5u	500m	2.9u
M9,M10	NMOS-Current	10	15u	1u	7.5 u

4) Cascode biasing

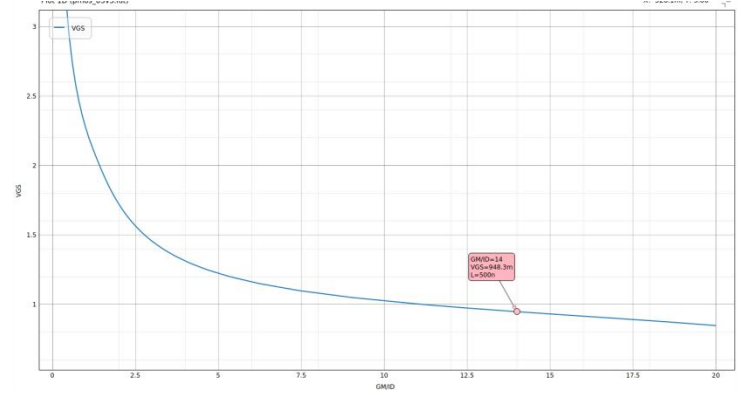
$$VCASCN \approx VGSN + V^*$$

$$VCASCN = 833\text{m} + 200\text{m} = 1.03\text{ V}$$

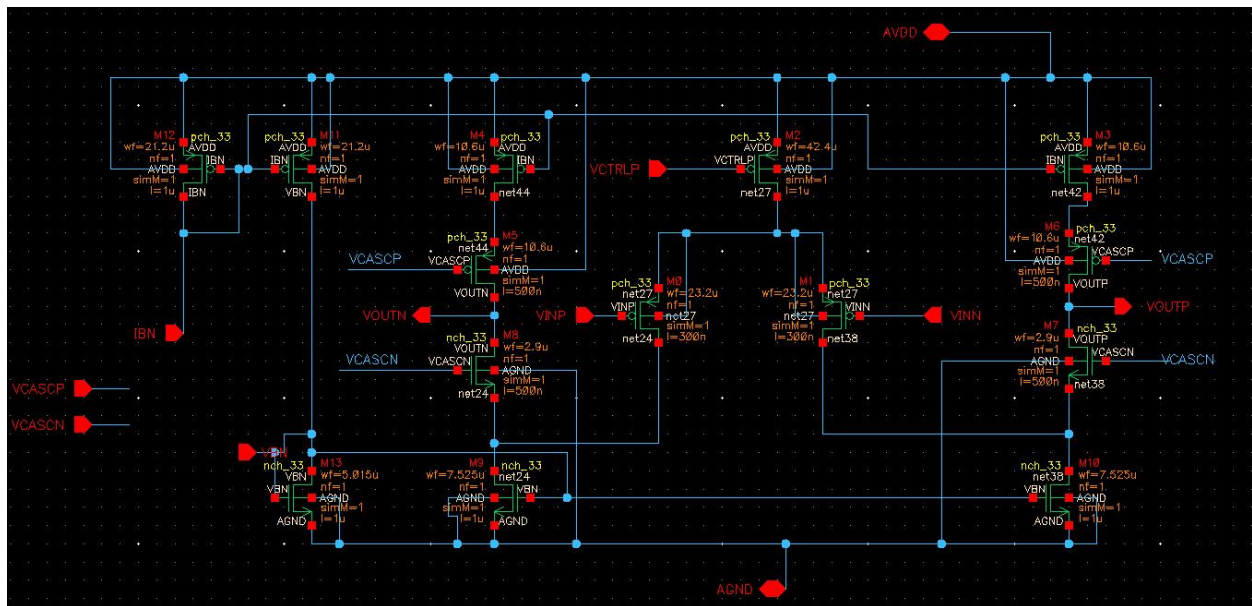


$$VCASCP \approx VDD - |V_{GSP}| - V^*$$

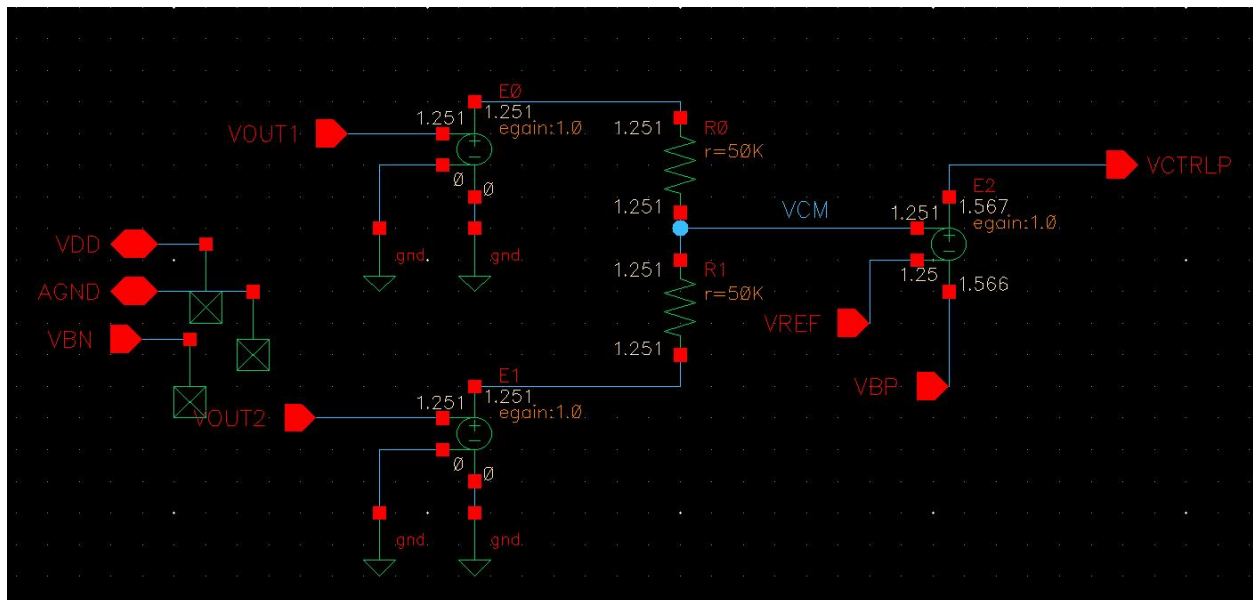
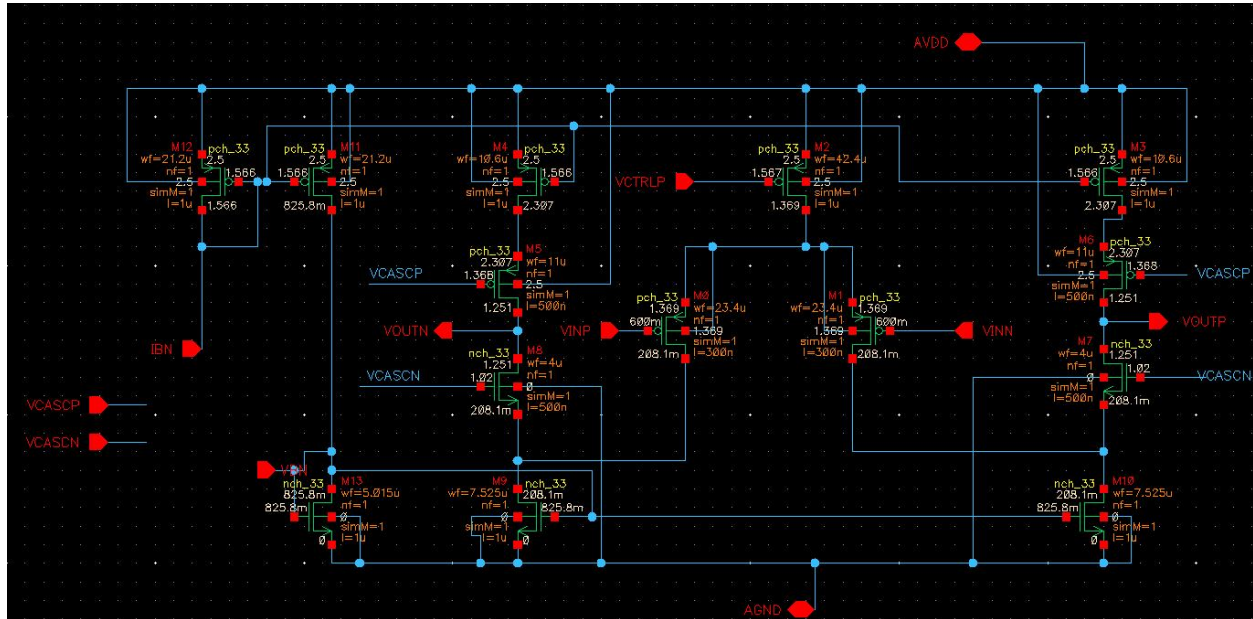
$$VCASCP = 2.5 - 948\text{m} - 200\text{m} = 1.352\text{ V}$$

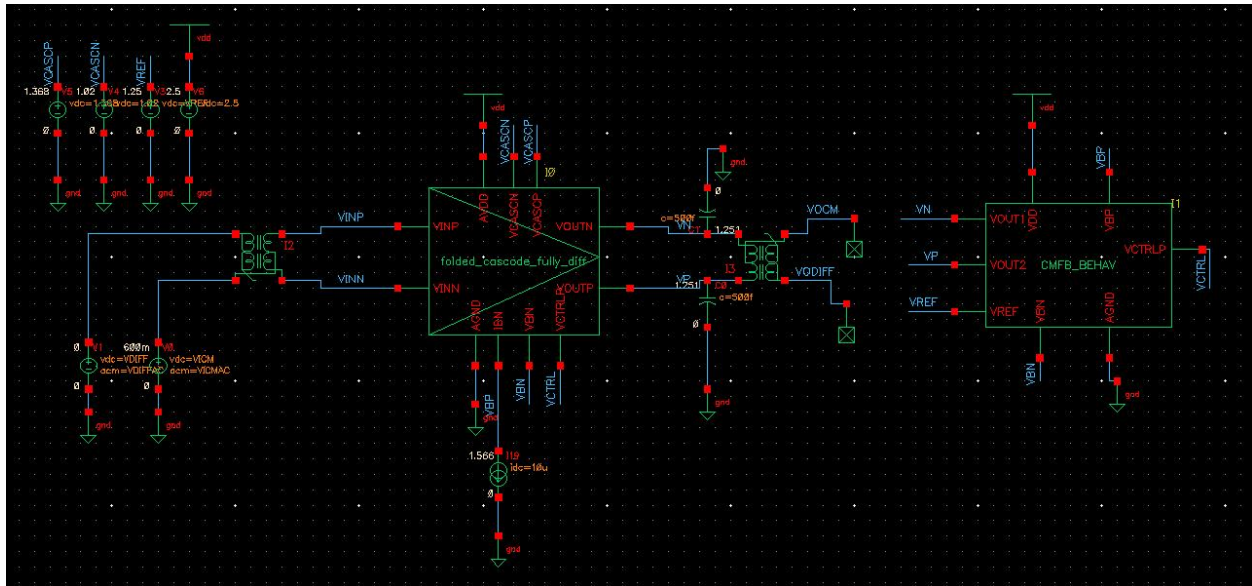


Final Schematic



Part 3 – Open-Loop OTA Simulation (Behavioral CMFB)





1) DC Operating Point

Transistor	Region	ID (uA)	Vov (V)	VDS (V)	VDSAT (V)	Gm/id
M0,M1	2	9.95u	24 m	1.1	92m	17
M2	2	19.89u	148m	1.1	148m	10.4
M3,M4	2	4.8u	149m	0.2	149m	10.2
M5,M6	2	4.8u	59m	1.05	108m	15.1
M7,M8	2	4.8u	38.39m	1.05	108m	15.1
M9,M10	2	17.75u	137m	0.204	156m	10.2
M11	2	10.14u	149.8m	1.607	149m	10.3
M12	2	10u	149.8m	0.933	149m	10.44
M13	2	10.14u	138.3m	0.825	165m	10.35

Q: What is the CM level at the OTA output?

1.25, CMFB network working on making cmfb equal to VREF

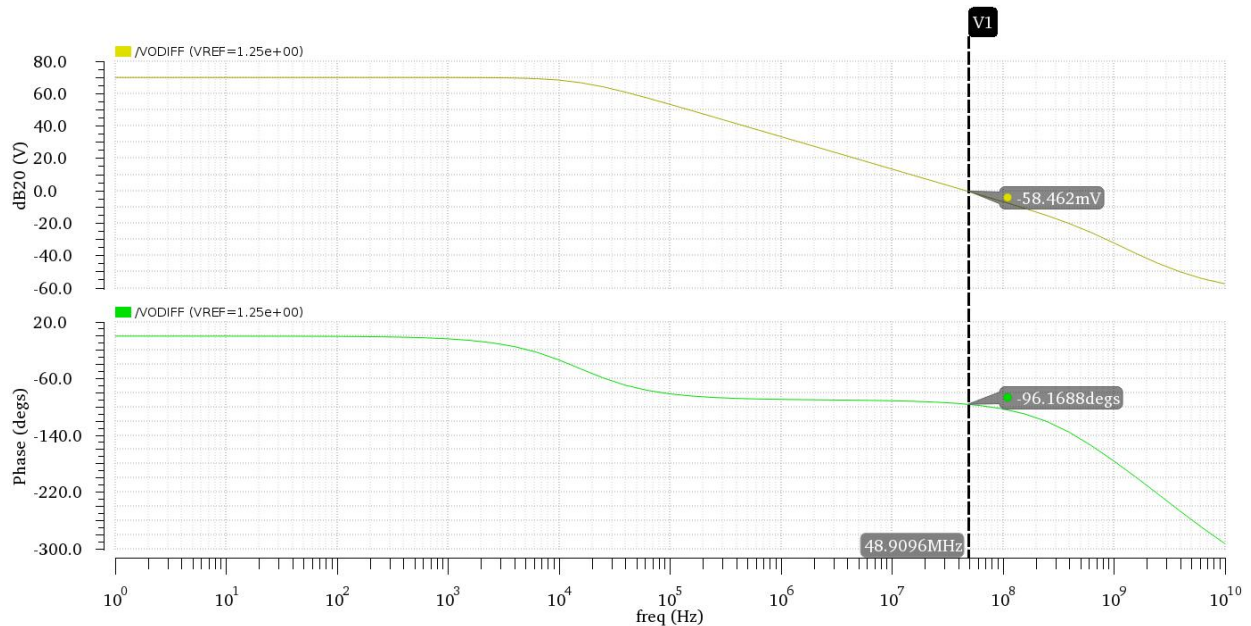
Q: What are the differential input and output voltages of the error amplifier, and what is the relation between them?

Diff input = $1.251 \text{ m} - 1.25 \text{ m} = 0.001 \text{ mV}$

Diff output VCTRL = $1.567 \text{ m} - 1.566 \text{ m} = .001 \text{ m}$

Gain = output / input = 1 V/V

2) Differential AC Response



my_labs:FOLDED_TB:1	A0	3.146k			
my_labs:FOLDED_TB:1	A0_db	69.95			
my_labs:FOLDED_TB:1	BW	15.12k			
my_labs:FOLDED_TB:1	UGF	49.86M			
my_labs:FOLDED_TB:1	GBW	47.55M			
my_labs:FOLDED_TB:1	phaseMargin(VF("/VODIFF"))	83.94			

Hand analysis

DC gain = $GM_{Rout} = g_{m1} [r_{o3} (g_{m3} (r_{o2} || r_{o1})) || [r_{o4} (g_{m5} (r_{o5}))]] = 3.1k$

$C_{out} = 500fF + C_{OTA_output}$

$BW = 1/2 * \pi * R_{out} C_{out} = 15.81kHz$

$GBW * UGF = 49Mhz$

$UGF = GBW$

To get inuruation for PM we know for crictal

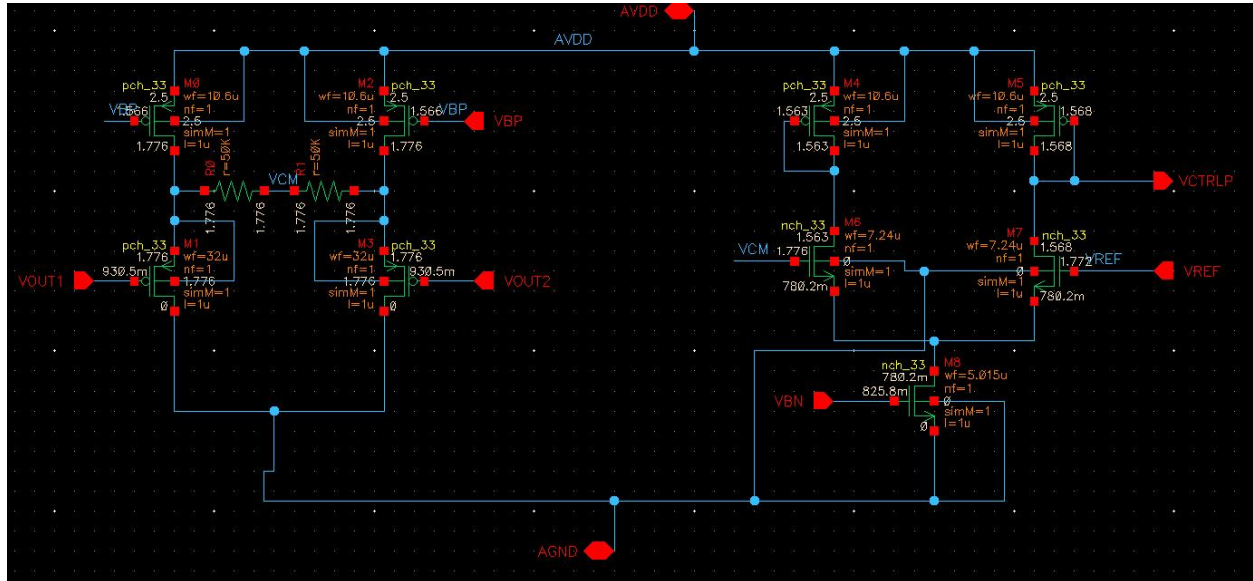
$Wu_LG/Wp2 = 1/4$, $Wu_ol /Wp2 = B * Wu_LG/Wp2 = 1/3 * 1/4 = 1/12$

$PM = 90 - \tan^{-1}(1/12) = 85.3$

Simulated vs Hand-Calculated Parameters

Parameter	Hand Calculation	Simulation
DC Gain (dB)	3.1 k (69.82 dB)	3.146k (69.95 dB)
BandWidth	15.81 kHz	15.1 khz
UGF	50 Mhz	49.8 Mhz
GBW	50 Mhz	47.9 Mhz
PM	85.3	83

Part 4 – Open-Loop OTA Simulation (Actual CMFB)



CMFB Circuit Design

Putting 20 u current for the entire cmfb network

For current source we use $gm/id = 15$ and $L = 1u$ same as in Folded ota from id/W chart we get W

For input pmos we use $gm/id = 10$ and $L = 1$ from id/W chart we get the W

Transistor	L (um)	gm/ID	ID (uA)	W (um)
M0,M2,M4,M5 PMOS	1u	10	5u	10.6u
M1,M3 PMOS	1u	15	5u	32u
M6,M7 NMOS	1u	15	5u	7.24u

We get V_{GS} for input pmos so input for error will be $V_{GSP} + V_{OCM}$

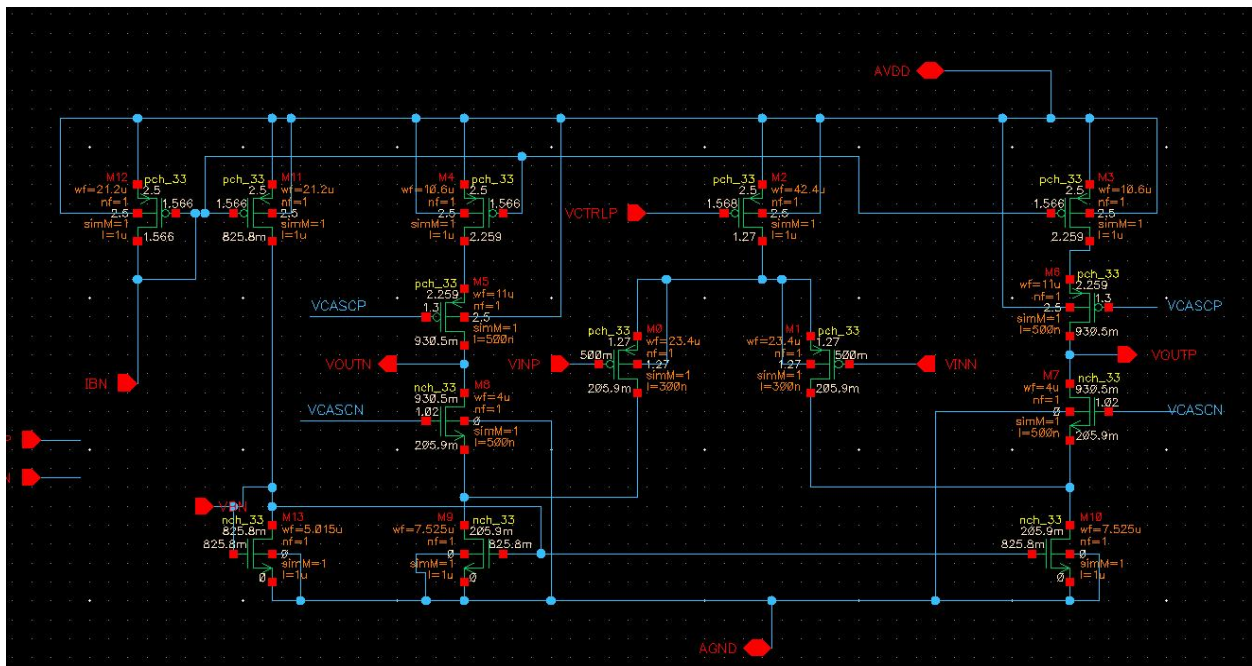
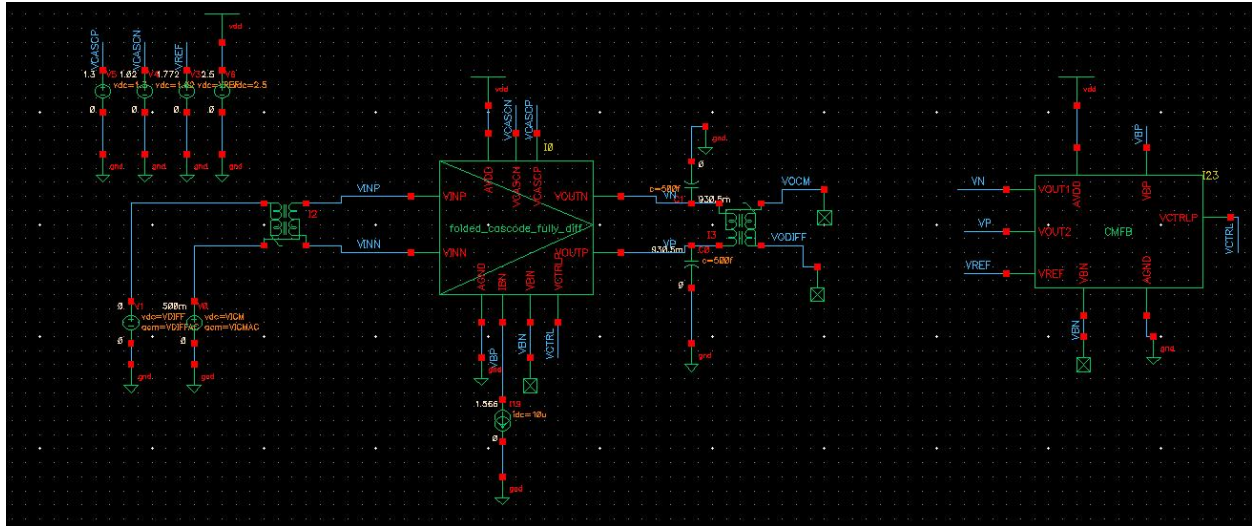
$V_{GSP} = 846$ mV for input pmos

maximum swing will be determine from error not OTA equal to $V_{DD} - V^* - V_{GSP} = 1.454$

and minimum swing will be $2 * V^* = 400$ m

to get maximum swing V_{OCM} must be at 925mV

so vref will equal to $V_{OCM} + V_{GSP} = 1.771V$



Transistor	Region	ID (uA)	Vov (V)	VDS (V)	VDSAT (V)	Gm/id
M0,M1	2	9.95u	24 m	1.1	92m	17
M2	2	19.89u	148m	1.1	148m	10.4
M3,M4	2	4.8u	149m	0.2	149m	10.2
M5,M6	2	4.8u	59m	1.05	108m	15.1
M7,M8	2	4.8u	38.39m	1.05	108m	15.1
M9,M10	2	17.75u	137m	0.204	156m	10.2
M11	2	10.14u	149.8m	1.607	149m	10.3
M12	2	10u	149.8m	0.933	149m	10.44
M13	2	10.14u	138.3m	0.825	165m	10.35

For CMFB NETWORK

Transistor	Region	ID (uA)	Vov (V)	VDS (V)	VDSAT (V)	Gm/id	gm (uS)
M0,M2	2	4.98u	150 m	754.5m	150m	10.1	51u
M1,M3	2	4.98u	61m	1.76	94m	15.1	75u
M4	2	5.1u	153m	937m	151m	10.2	53u
M5	2	4.9u	148.3m	932m	151m	10.3	51u
M6	2	5.1u	55m	782.4m	115.3	14.9	77u
M7	2	4.9u	51m	787.6m	113m	15.1	74u
M8	2	10.12u	135m	780m	165m	10.3	104.9

Q: What is the CM level at the OTA output? Why?

930m is it close to VREF - VGSP , CMFB network working on making cmfb equal to VERF

Q: What are the differential input and output voltages of the error amplifier? What is the relation between them?

Diff input = 1.776 m - 1.772m = 0.004mV

Diff output VCTRL = 1.568m - 1.565m = .003mV

Gain = output / input = 1.33 V/V is the gain of error amp

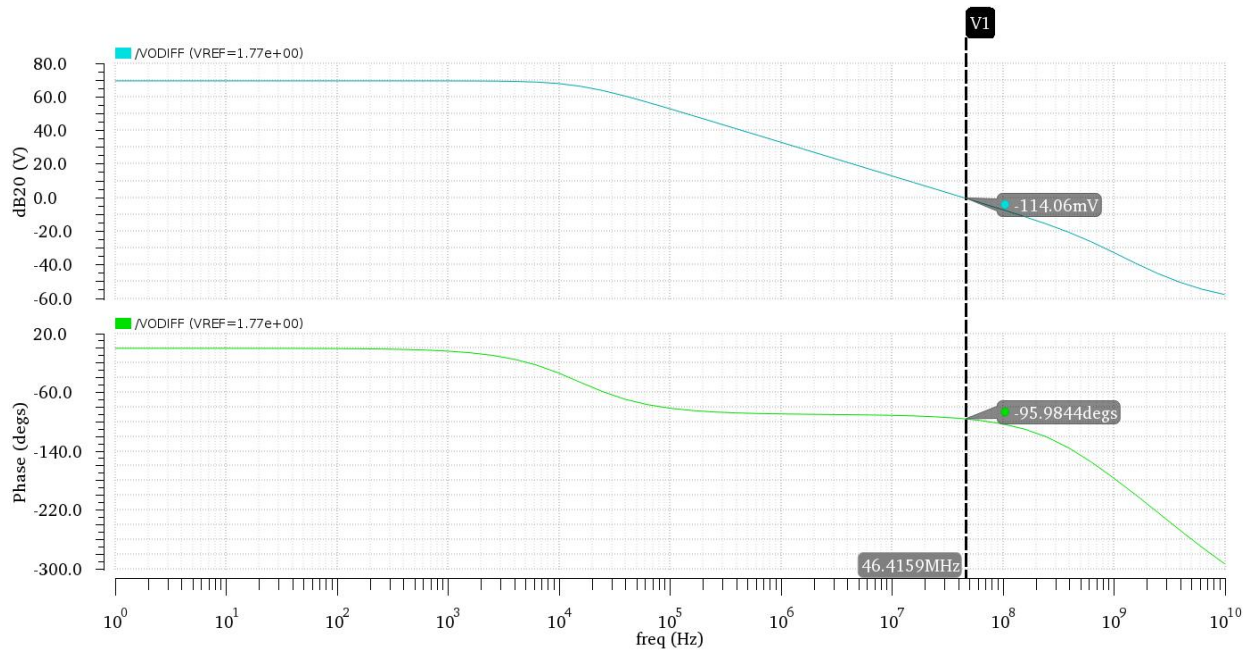
$AV_{CMFB} = gm_7 * 1 / gm_5 = 1.45$

$V_{out,max} = VDD - (VGSP) - VDS_{sat3} - vdsat2 = 1.45$

$V_{out,min} = vdsat7 + vdsat8 = 0.25V$

Output Swing = 1.2

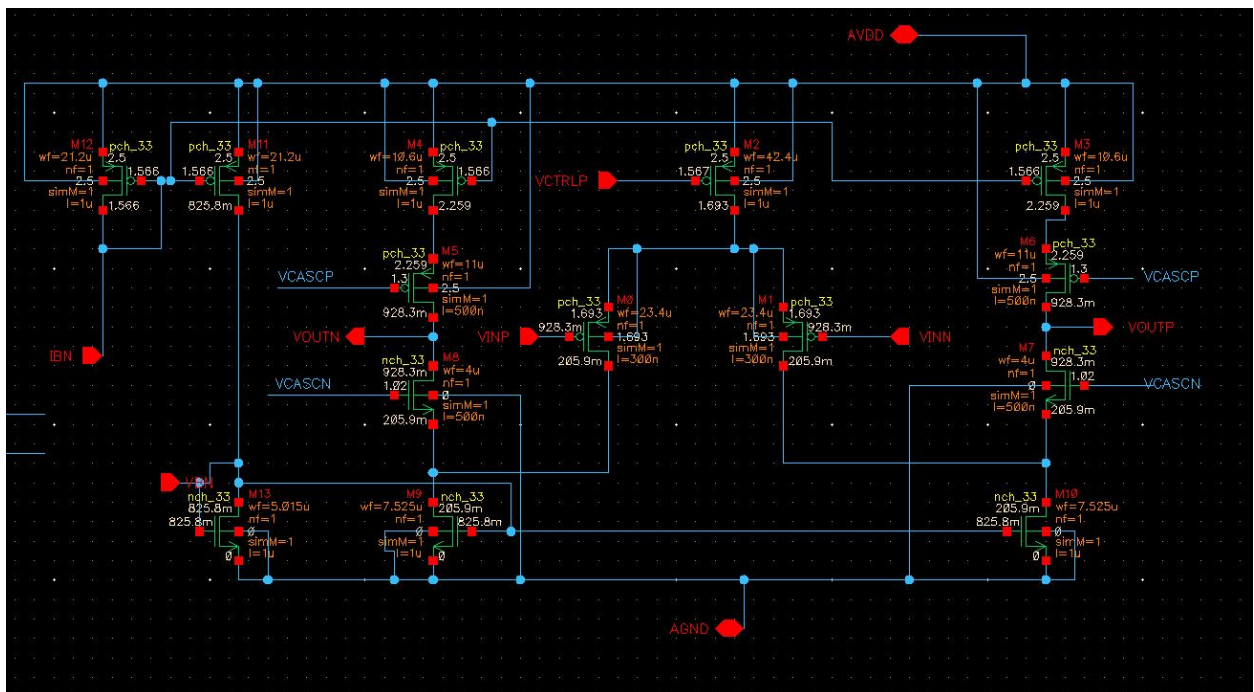
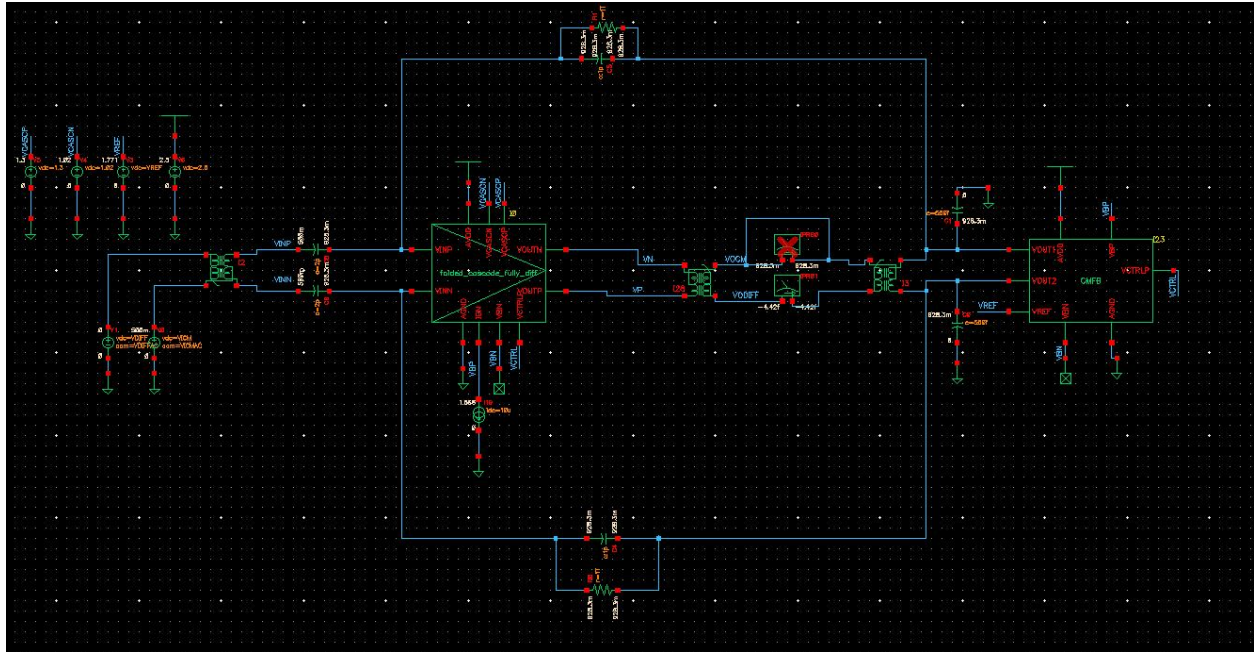
2) Differential AC Response (Actual CMFB)

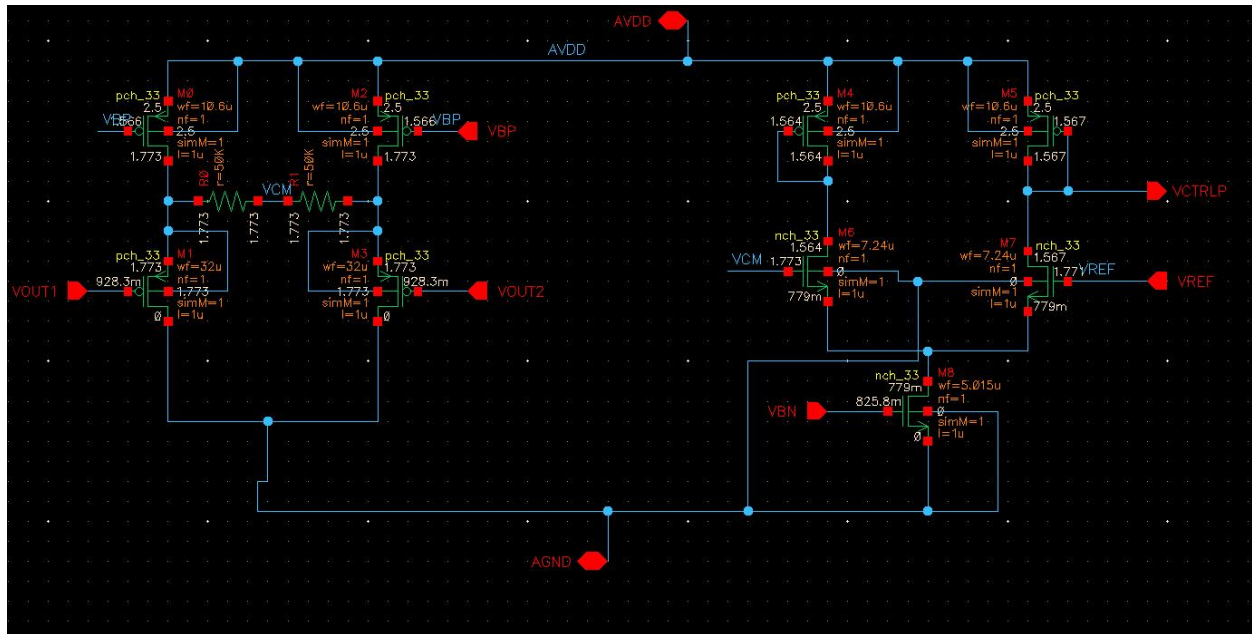


my_labs:FOLDED_TB_actual:1	A0	3.002k			
my_labs:FOLDED_TB_actual:1	A0_dB	69.55			
my_labs:FOLDED_TB_actual:1	BW	14.99k			
my_labs:FOLDED_TB_actual:1	UGF	46.82M			
my_labs:FOLDED_TB_actual:1	GBW	45.01M			
my_labs:FOLDED_TB_actual:1	PM	84.23			

Part 5 – Closed-Loop Simulation (AC and STB Analysis)

1) DC Operating Point (Closed Loop)

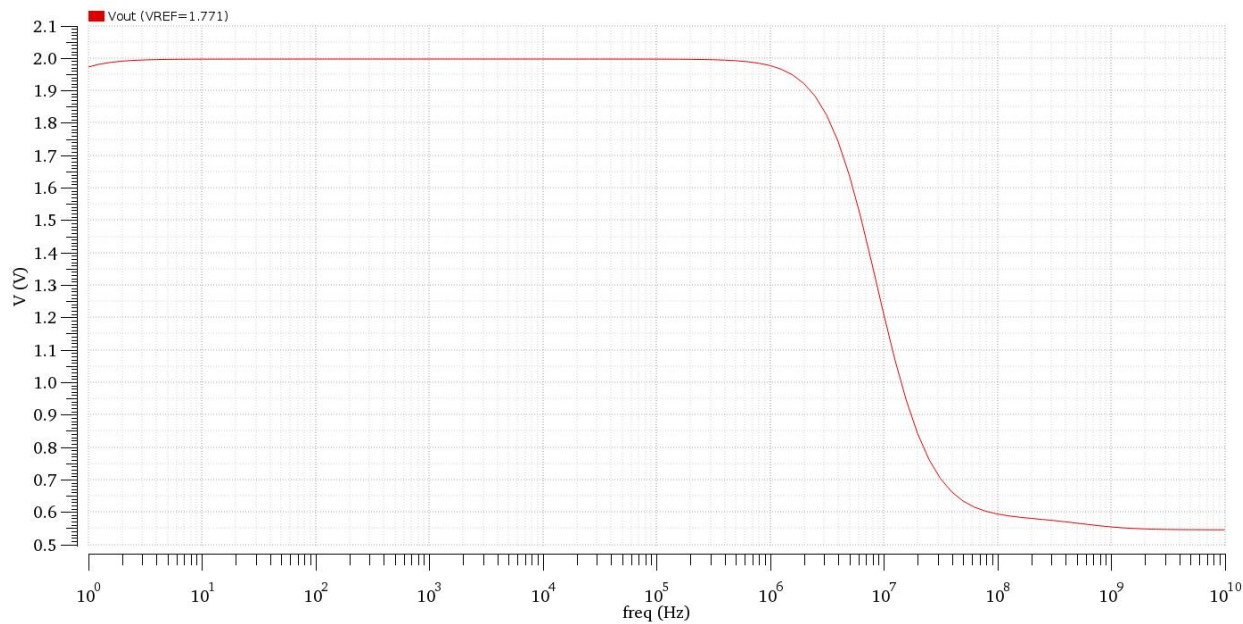




my_labs:FOLDED_TB_actual_part5:1	Vocm	928.3m
my_labs:FOLDED_TB_actual_part5:1	Vicm_	928.3m

the output cm level = 928.3m and the input cm level is 928.3m the feed resistor parallel to C_f is making v_{outcm} to bias the input voltage .

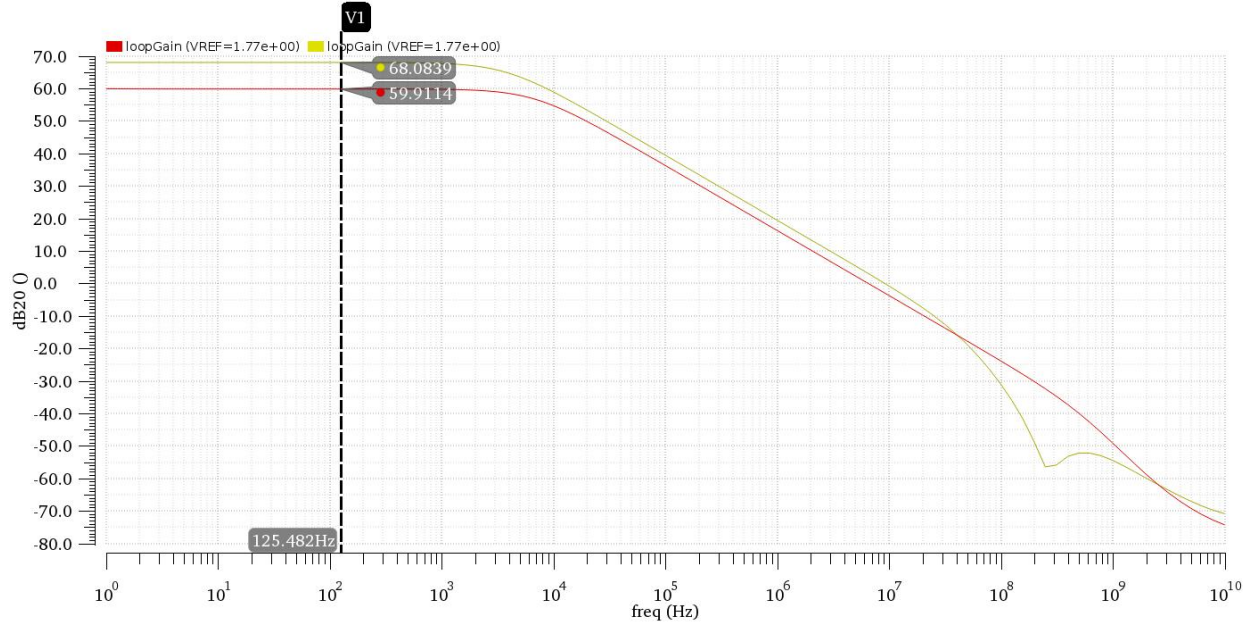
2) Differential Closed-Loop AC Response



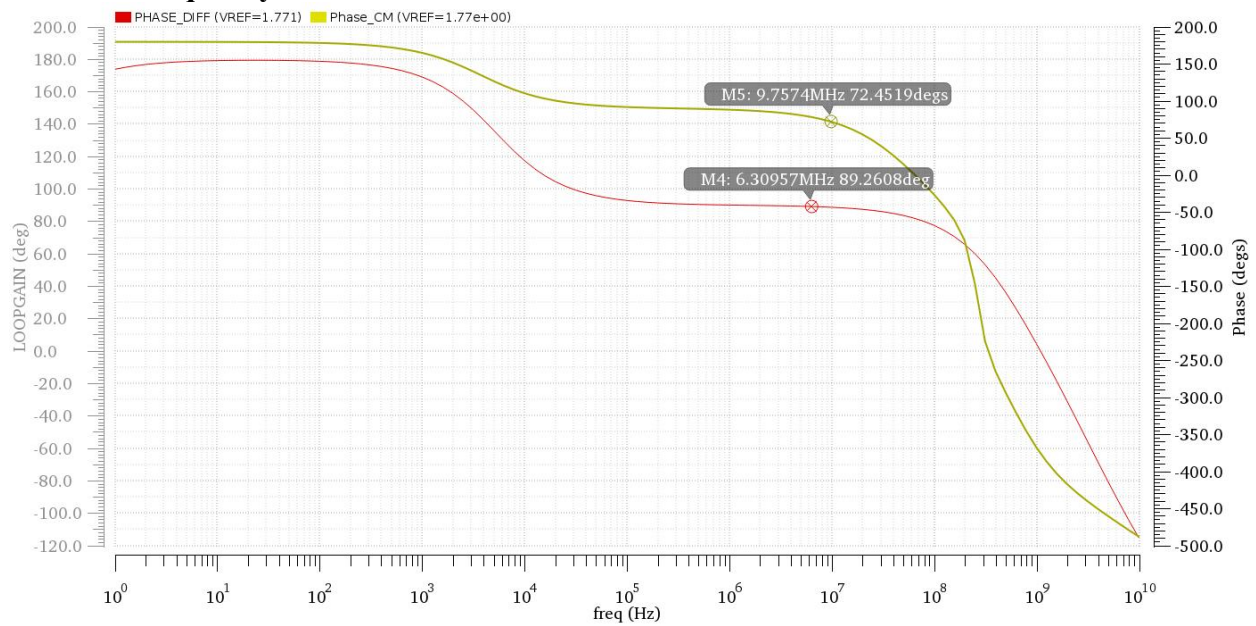
my_labs:FOLDED_TB_actual_part5:1	A0	1.998			
my_labs:FOLDED_TB_actual_part5:1	A0_dB	6.012			
my_labs:FOLDED_TB_actual_part5:1	BW	7.586M			
my_labs:FOLDED_TB_actual_part5:1	GBW	15.01M			
my_labs:FOLDED_TB_actual_part5:1	UGF	14.47M			

3) Differential and CMFB Loop Stability (STB Analysis)

LoopGain (dB) VS frequency



Phase VS frequency



For Diff loop gain

my_labs:FOLDED_TB_actual_part5:1	PM	89.23
my_labs:FOLDED_TB_actual_part5:1	GBw	6.556M
my_labs:FOLDED_TB_actual_part5:1	bandwidth(mag(getData("loopGain" ?result "stb")) ...	6.531k
my_labs:FOLDED_TB_actual_part5:1	LG_dB	60.01

For Cm loop gain

my_labs:FOLDED_TB_actual_part5:1	PM	73.26
my_labs:FOLDED_TB_actual_part5:1	GBw	9.548M
my_labs:FOLDED_TB_actual_part5:1	bandwidth(mag(getData("loo...	3.754k
my_labs:FOLDED_TB_actual_part5:1	LG_dB	68.09

Loop	DC Loop Gain (dB)	GBW	Phase Margin
Differential	62.48	6.784 M	89.19
Common mode	68.09	9.548 M	73.26

Both CM loop & Diff Loop has a higher GBW and high phase margin because we need it to be fast to reset the common_mode level of amplifier

Both CM loop & Diff Loop have high phase margin because it like a first order system

Compare DC LG and GBW of the diff loop with those obtained from open-loop simulation

$$\text{Dc loop gain} = \beta * A_{ol} = 1/3 * 3.164k = 60.46 \text{ dB}$$

$$\text{GBW}_{LG} = \text{GBW}_{ol} * \beta * c_l / c_{out} = 45M * 1/3 * 0.5 = 7.5 \text{ MHz}$$

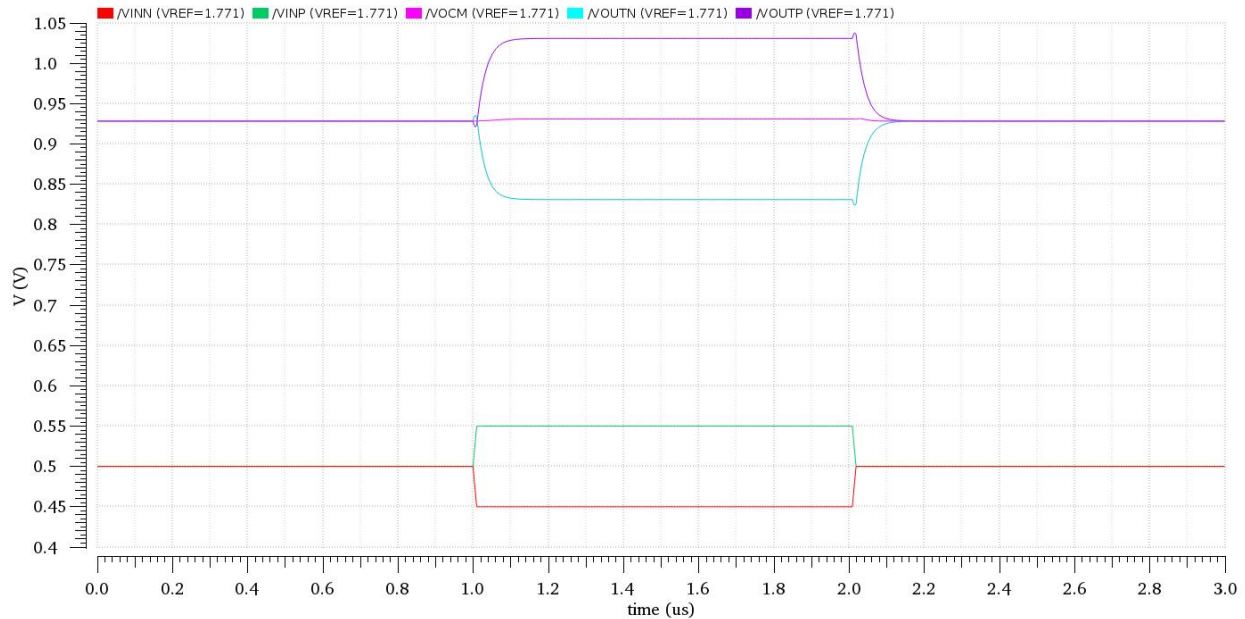
Loop	DC Loop Gain (dB)		GBW	
	Open loop simulation	STB simulation	Open loop simulation	STB simulation
Diff loop	60.46 dB	60.1 dB	7.5 MHz	6.5 Mhz

Ther error is due to β is not ecactly 1/3 but it equal to $c_f / (c_f + c_{in} + c_{ota})$

Part 6 – Closed-Loop Simulation (Transient Analysis)

1) Differential Input Pulse

VINP, VINN, VOUTP, VOUTN, and VOVM



There is no any ringing or overshoot due to PM or due to pole damping behavior since the PM is greater than 76.

the spikes we see in plot is due to high frequency beyond closed loop bandwidth making oscillation near to the nondominant pole.

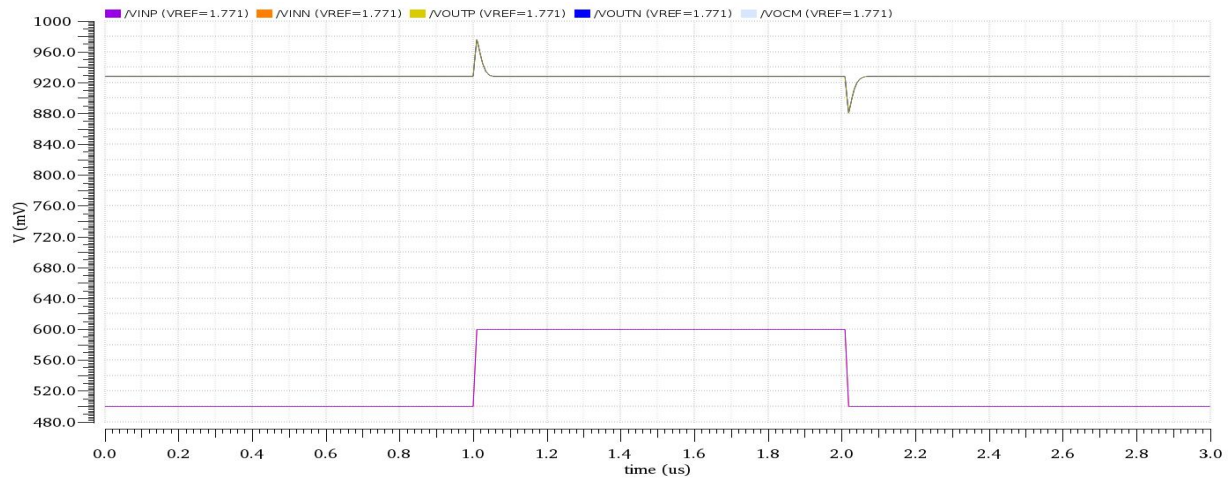
my_labs:FOLDED_TB_actual_part6:1	Settling time	94.27n
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Settling time is 94.27 n is less than 100n (specs required)

To decrease settling time we can increase gm/id for input pair to increase GBW .

3) Common-Mode Input Pulse

VINP, VINN, VOUTP, VOUTN, and VOCM

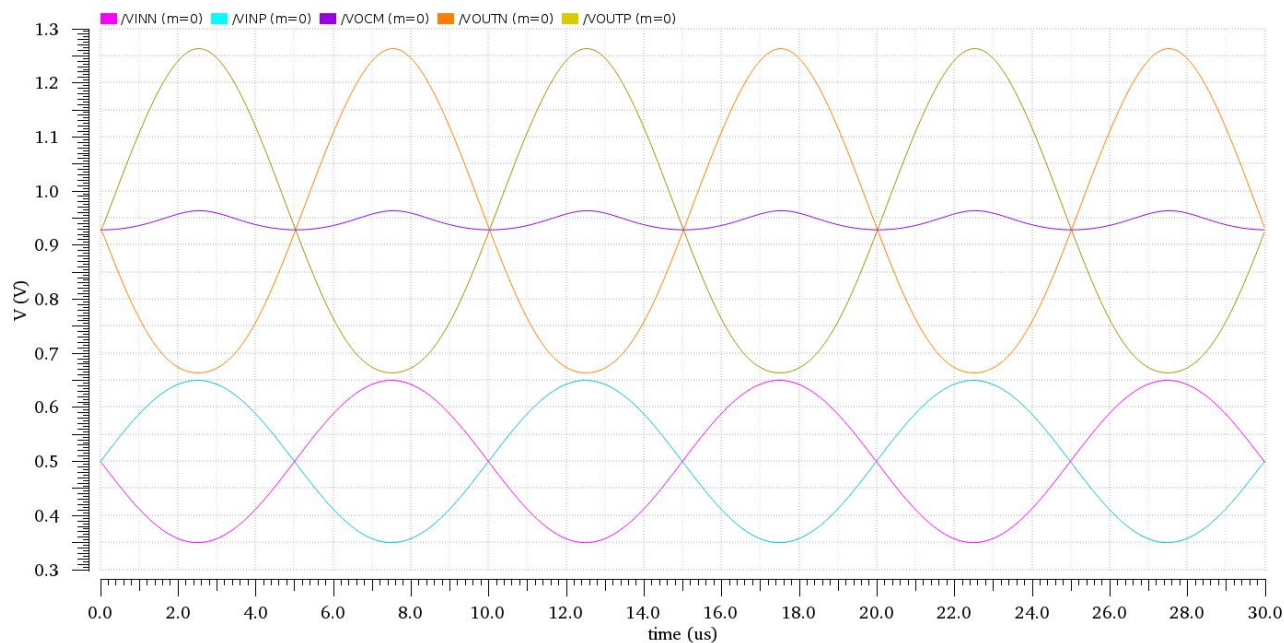


There is no any ringing or overshoot due to PM or due to pole damping behavior since the PM is greater than 76.

the spikes we see in plot is due to high frequency beyond closed loop bandwidth making oscliation near to the nondominant pole.

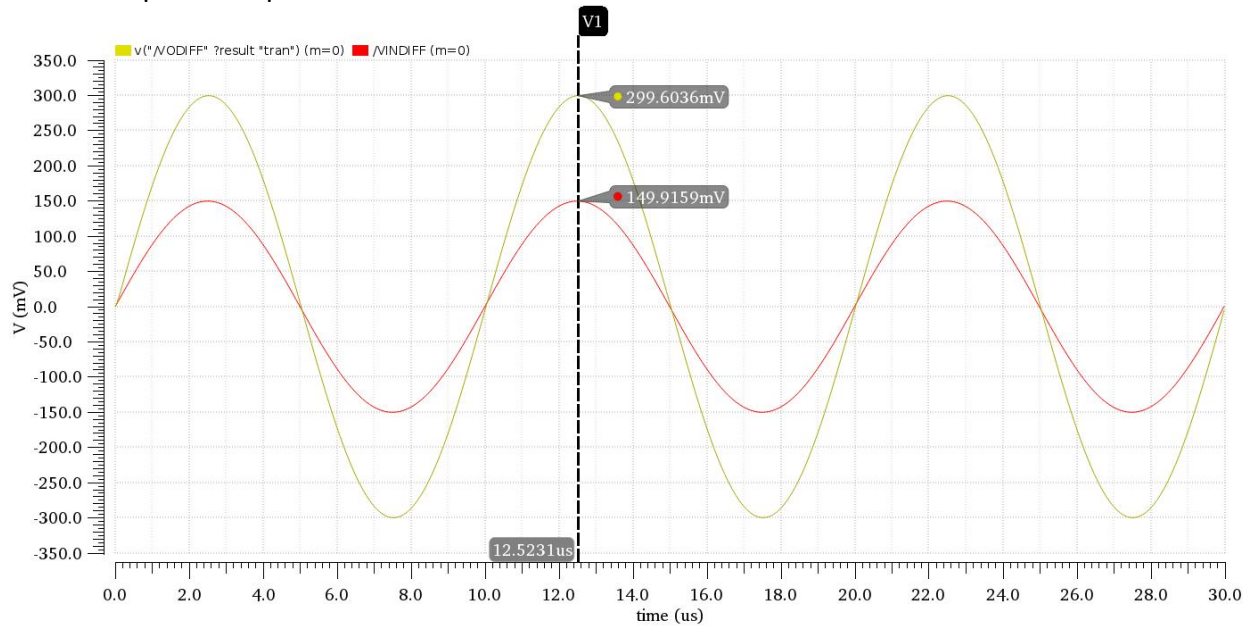
4) Output Swing (Sinusoidal Input)

VINP, VINN, VOUTP, VOUTN, and VOCM



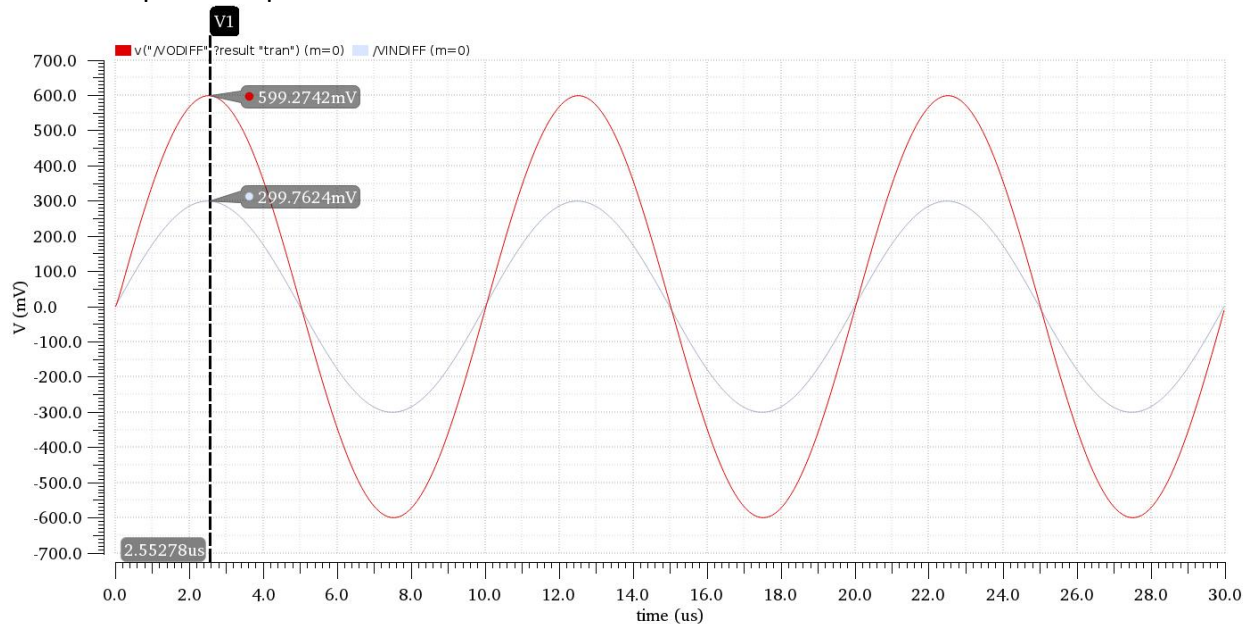
VIDIFF and VODIFF overlaid

150mV amplitude input



my_labs:FOLDED_TB_actual_part6:1	peakToPeak(v("/VODIFF" ?res...	599.3m			
my_labs:FOLDED_TB_actual_part6:1	peakToPeak(v("/VIDIFF" ?re...	300m			
my_labs:FOLDED_TB_actual_part6:1	gain	1.998			

300mV amplitude input



my_labs:FOLDED_TB_actual_part6:1	peakToPeak(v(\"/VODIFF\" ?res...	1.199
my_labs:FOLDED_TB_actual_part6:1	peakToPeak(v(\"/VINDIFF\" ?re...	599.9m
my_labs:FOLDED_TB_actual_part6:1	gain	1.998

Output Swing 1.2VP-P is achieved as specs

- Magic battery for CascodN

M1 and M0 form the magic battery that will be used to generate the cascode bias voltage

M1 is always in saturation

M0 is always in triode ,

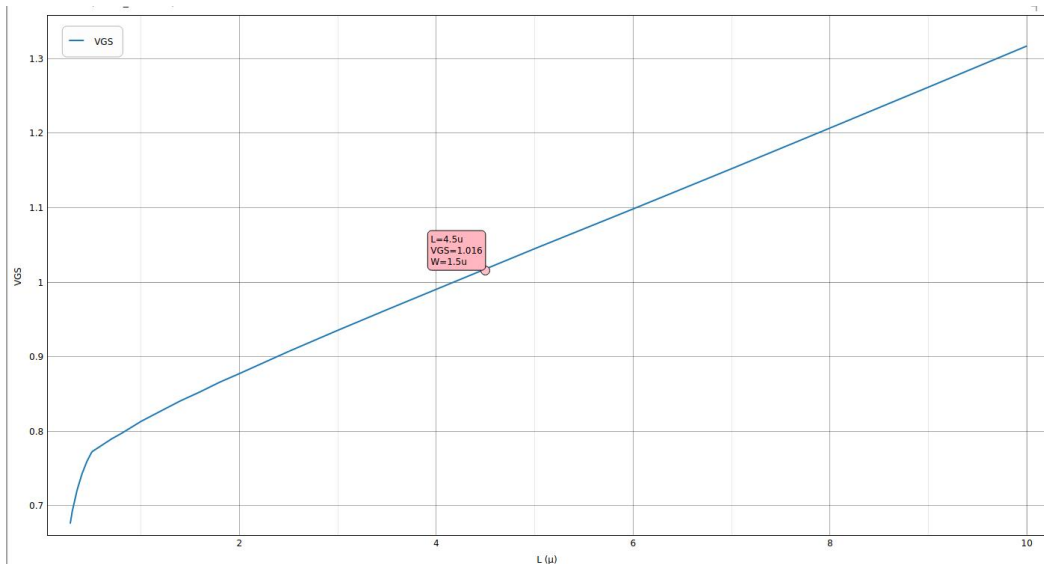
We want v_{gs} of M0 = $V_B = 1.02$ V

M2 same sizing for pmos current mirror

$W = 5.3\mu$ for current $2.5\mu A$

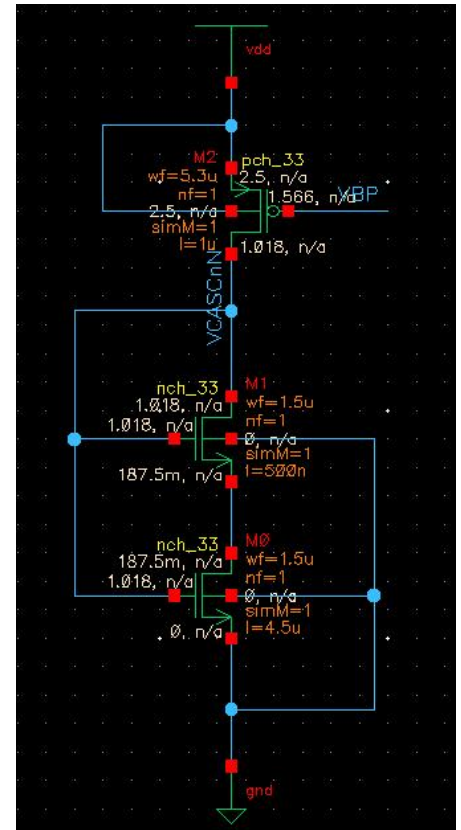
For M1 use the same bias point and sizing for Cascdn transistor in ota

For M0 same W for M1 and sweep L to get a specific v_{gs} from below graph L for M0 = 4.5μ



my_labs:FOLDED_TB_actual_part5:1	v("/VCASCnN" ?result "dcOp")	1.037
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my_labs:FOLDED_TB_actual_part5:1	getData("M0.m0:id" ?result "dcOp")	2.525u
my_labs:FOLDED_TB_actual_part5:1	getData("M1.m0:id" ?result "dcOp")	2.525u
my_labs:FOLDED_TB_actual_part5:1	getData("M2.m0:id" ?result "dcOp")	-2.525u



- Magic battery for CascodP

M8 and M7 form the magic battery that will be used to generate the cascode bias voltage is 1.3 V

M7 is always in saturation

M8 is always in triode ,

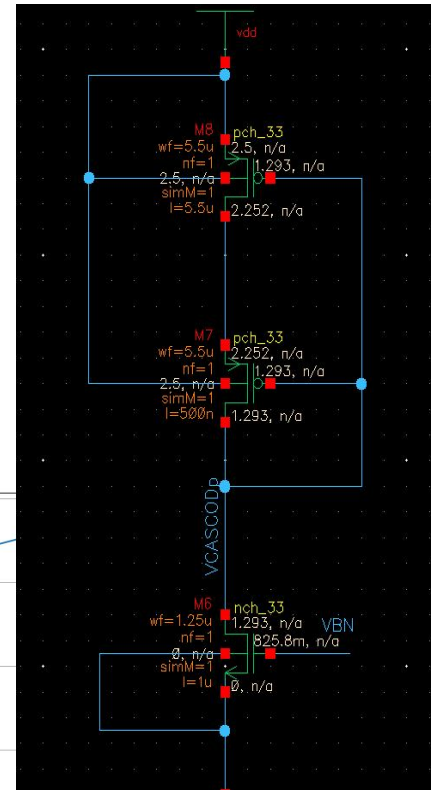
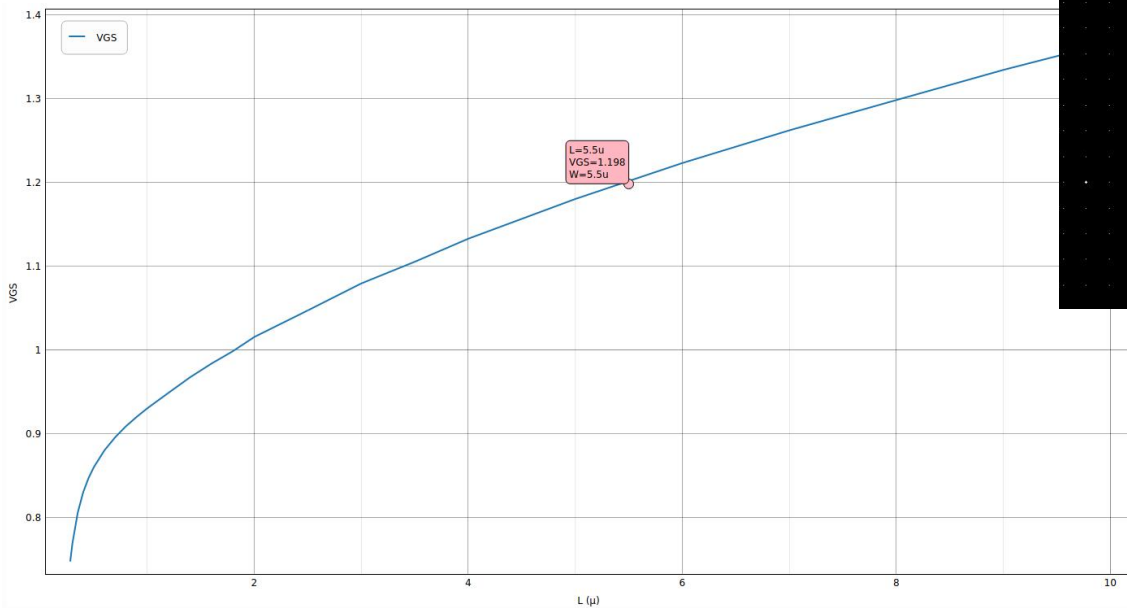
We want $v_{gs_M8} = 2.5 - 1.3 = 1.2$

M6 same sizing for Nmos current mirror

$W = 1.25\mu$ for current 2.5 μA

For M7 use the same bias point and sizing for CascodP transistor in ota

For M8 same W for M7 and sweep L to get a specific v_{gs} from below graph L for M8 = 5.5u



my_labs:FOLDED_TB_actual_part5:1	v("/VCASCODp" ?result "dcOp")	1.293
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my_labs:FOLDED_TB_actual_part5:1	getData("M6.m0:id" ?result "...")	2.342u
my_labs:FOLDED_TB_actual_part5:1	getData("M7.m0:id" ?result "...")	-2.342u
my_labs:FOLDED_TB_actual_part5:1	getData("M8.m0:id" ?result "...")	-2.342u

Closed loop Simulation with Vcascode implementaions

my_labs:FOLDED_TB_actual_part5:1	A0	1.998			
my_labs:FOLDED_TB_actual_part5:1	A0_dB	6.014			
my_labs:FOLDED_TB_actual_part5:1	BW	7.786M			
my_labs:FOLDED_TB_actual_part5:1	GBW	15.4M			
my_labs:FOLDED_TB_actual_part5:1	UGF	14.86M			

$$Fom = GBW / (\text{total current} * \text{total area})$$

$$\text{Total current} = 70 \text{ uA}$$

$$\text{Total area} = 224 \text{ um}^2$$

$$GBW_{cl} = 15\text{MHz}$$

$$FOM = 9.576 \times 10^{-4} \text{ MHz}/(\mu\text{A} \cdot \mu\text{m}^2) = 9.576 \times 10^{20} \text{ Hz}/(\text{A} \cdot \text{m}^2)$$